Writing results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Keep counts, outcome definitions and failed conditions visible without substituting unmeasured benefits.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Writing the methods section

> **My remembered idea, example and correction**

Negative results and claim revision

> **My remembered idea, example and correction**

Running the primary experiment

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Results report actual counts, denominators, measured outcomes and descriptive magnitude before interpretation. Main A passes 8/12 ≈ 66.667% and B 9/12 = 75%, gain one of twelve or about 8.333 percentage points. Three B failures involve conflicting sources. Their total slice denominator is unknown, so no conflict-slice rate can be inferred. No user study occurred.

  Separate picture measures answer support on twelve fixed questions from one approved collection: A 9/12 = 75%, B 10/12 ≈ 83.333%, again one supported-answer gain. These are distinct scenarios and outcomes. Support is not learner improvement; a claim every student becomes smarter substitutes an unmeasured outcome and population. No timings, significance or stable population effect are supplied.

  Preserve failed conditions and the original primary result, and identify precisely what evidence a broader question would need. A limitation does not turn an unsupported claim into a supported one.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Results report actual counts, denominators, measured outcomes and descriptive magnitude before interpretation. Main A passes 8/12 ≈ 66.667% and B 9/12 = 75%, gain one of twelve or about 8.333 percentage points. Three B failures involve conflicting sources. Their total slice denominator is unknown, so no conflict-slice rate can be inferred. No user study occurred.

  Separate picture measures answer support on twelve fixed questions from one approved collection: A 9/12 = 75%, B 10/12 ≈ 83.333%, again one supported-answer gain. These are distinct scenarios and outcomes. Support is not learner improvement; a claim every student becomes smarter substitutes an unmeasured outcome and population. No timings, significance or stable population effect are supplied.

  Preserve failed conditions and the original primary result, and identify precisely what evidence a broader question would need. A limitation does not turn an unsupported claim into a supported one.

- B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred. Show the intermediate steps and give the conclusion with its scope.

Main results component	Supported statement
Observed pass rates	A 8/12 ≈ 66.667%; B 9/12 = 75%
Descriptive difference	1/12 ≈ 8.333 percentage points in B’s favor
Failure category	Three B failures involve conflicting sources
What category statement does not imply	Not that every conflict case failed; total conflict cases unknown
User-study status	No study occurred; classroom usefulness not measured
Paired identities / statistical effect	Not supplied; no reconstructed transitions or significance

Main results component	Supported statement
Scoped report	B passed one more of twelve supplied cases, with conflict failures retained

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Results report actual counts, denominators, measured outcomes and descriptive magnitude before interpretation. Main A passes 8/12 ≈ 66.667% and B 9/12 = 75%, gain one of twelve or about 8.333 percentage points. Three B failures involve conflicting sources. Their total slice denominator is unknown, so no conflict-slice rate can be inferred. No user study occurred.

  Separate picture measures answer support on twelve fixed questions from one approved collection: A 9/12 = 75%, B 10/12 ≈ 83.333%, again one supported-answer gain. These are distinct scenarios and outcomes. Support is not learner improvement; a claim every student becomes smarter substitutes an unmeasured outcome and population. No timings, significance or stable population effect are supplied.

  Preserve failed conditions and the original primary result, and identify precisely what evidence a broader question would need. A limitation does not turn an unsupported claim into a supported one.

- A report omits this qualification: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

  Explain how omitting it could change a reader's **interpretation**, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Results report actual counts, denominators, measured outcomes and descriptive magnitude before interpretation. Main A passes 8/12 ≈ 66.667% and B 9/12 = 75%, gain one of twelve or about 8.333 percentage points. Three B failures involve conflicting sources. Their total slice denominator is unknown, so no conflict-slice rate can be inferred. No user study occurred.

  Separate picture measures answer support on twelve fixed questions from one approved collection: A 9/12 = 75%, B 10/12 ≈ 83.333%, again one supported-answer gain. These are distinct scenarios and outcomes. Support is not learner improvement; a claim every student becomes smarter substitutes an unmeasured outcome and population. No timings, significance or stable population effect are supplied.

  Preserve failed conditions and the original primary result, and identify precisely what evidence a broader question would need. A limitation does not turn an unsupported claim into a supported one.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Candidate 10/15, baseline 9/15; no timings measured. Write result and two limits.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled writing results and limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of writing results and limitations.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Writing results and limitations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Writing results and limitations. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Candidate 10/15, baseline 9/15; no timings measured. Write result and two limits.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

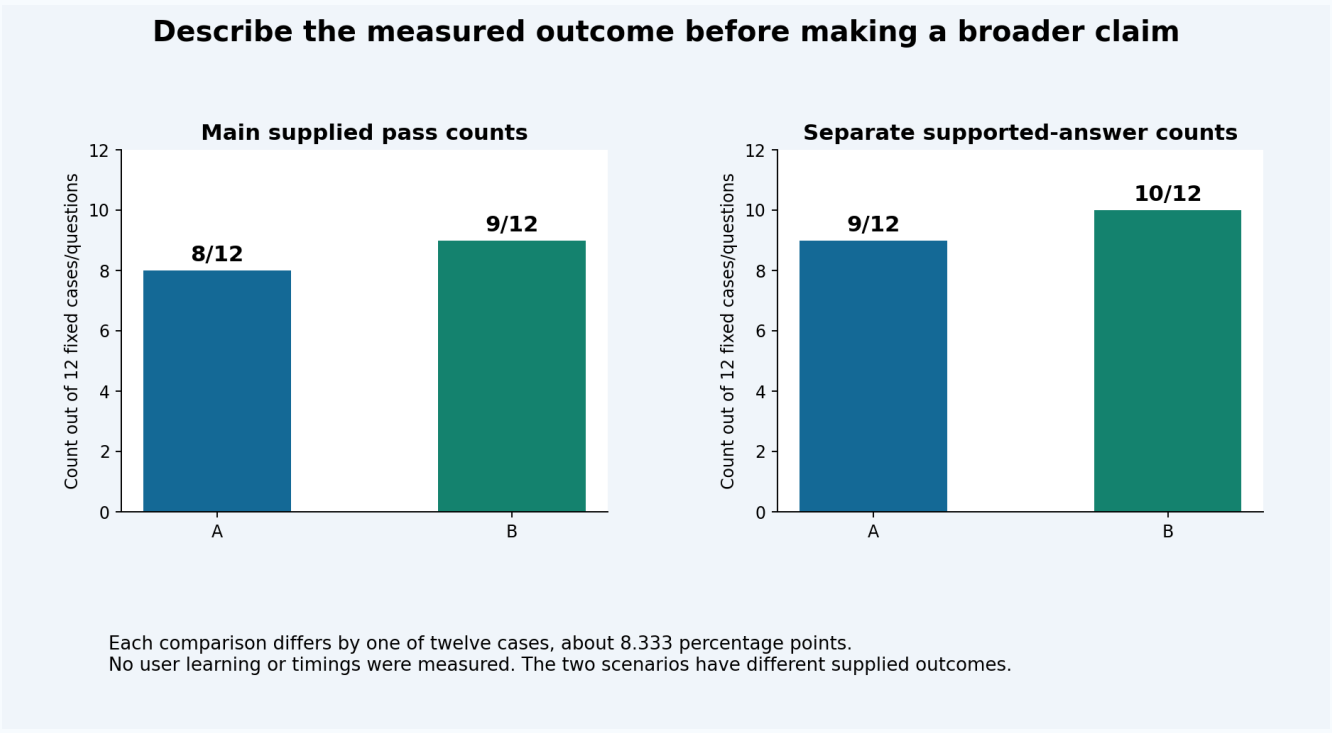
> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.
- A supports 9/12 answers; B supports 10/12. No user learning outcomes are recorded.
- Compute the supported-answer rates for 9/12 and 10/12. What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?
- Support used: alone / hint / worked example. This record is practice **evidence**.

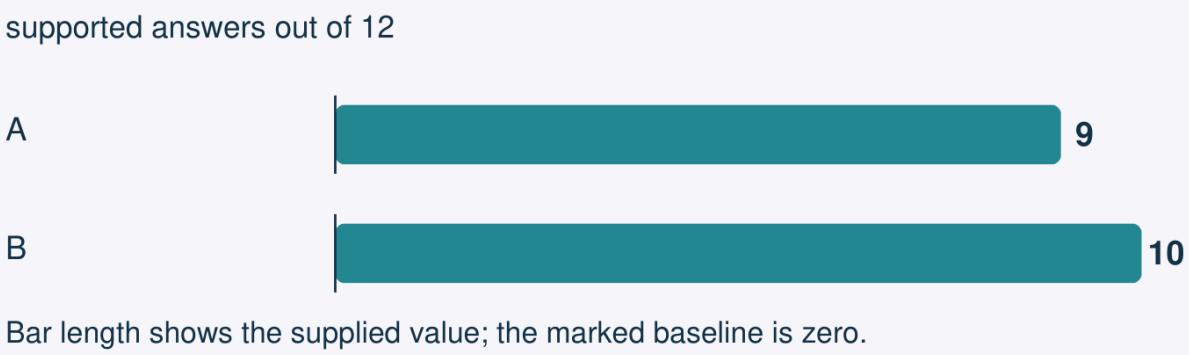
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/31

Supported answers in one collection



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/31

Writing results and limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Report the observation

A is 75%; B is about 83.3%, a one-case improvement.



2. Link to the method

Name the collection, rubric and twelve-case denominator.



3. State limits

Do not infer that children learn more from this answer-support measure.

CHANGE ONE THING

Write "B makes every student smarter".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.
- Which labelled step matches this detail? Do not infer that children learn more from this answer-support measure.
- 1. Report the **observation**
- 2. Link to the method
- 3. State limits
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: State limits Do not infer that children learn more from this answer-support measure.

Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

Changed-case question and optional hint

- Changed-case practice: Write "B makes every student smarter". Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Remove that sentence as unsupported by the study.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Remove that sentence as unsupported by the study. The measured unit was answers, not student understanding or intelligence.

Repair and retain the record

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

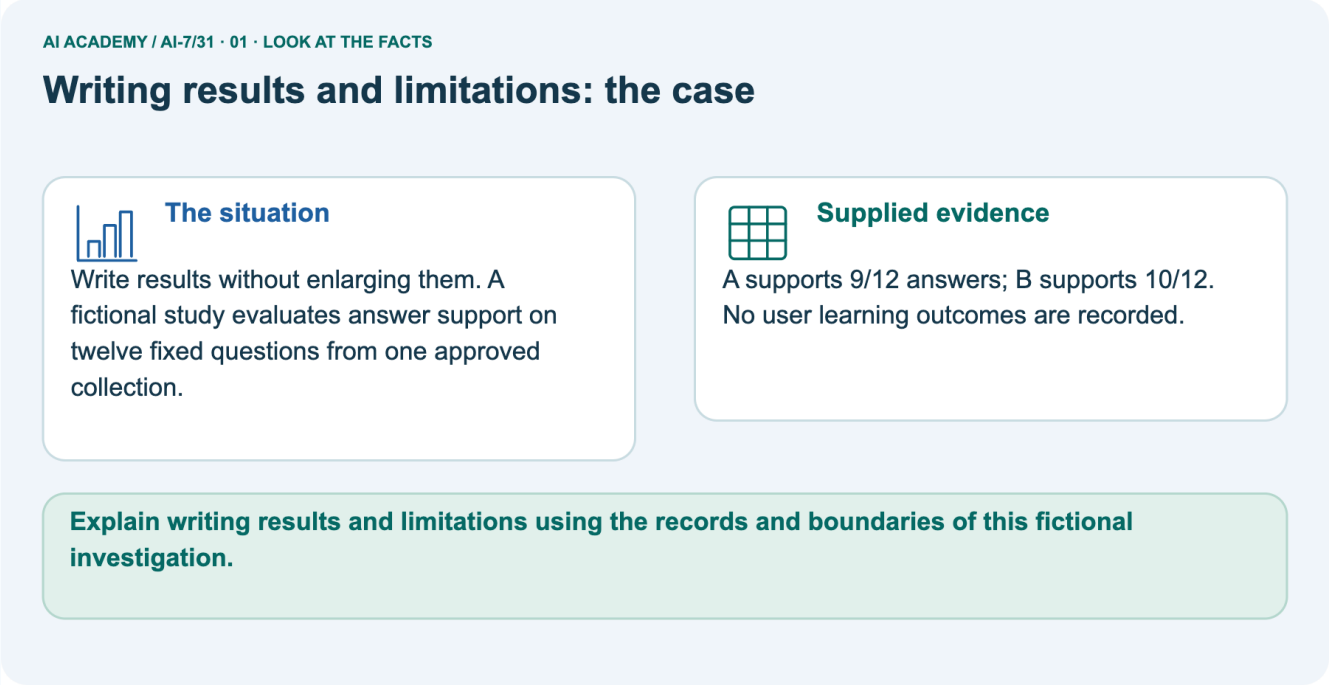
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Writing results and limitations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

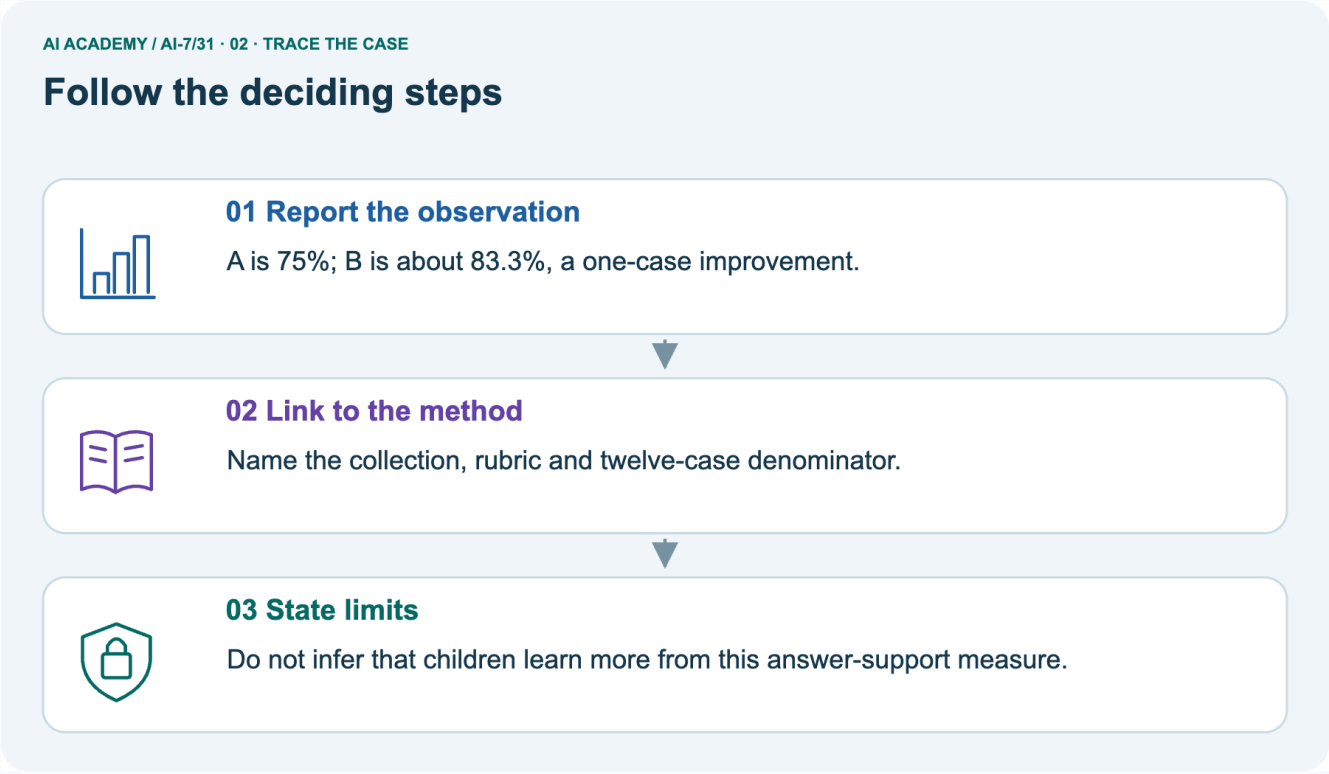
Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection. Supplied facts: A supports 9/12 answers; B supports 10/12.

No user learning outcomes are recorded.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



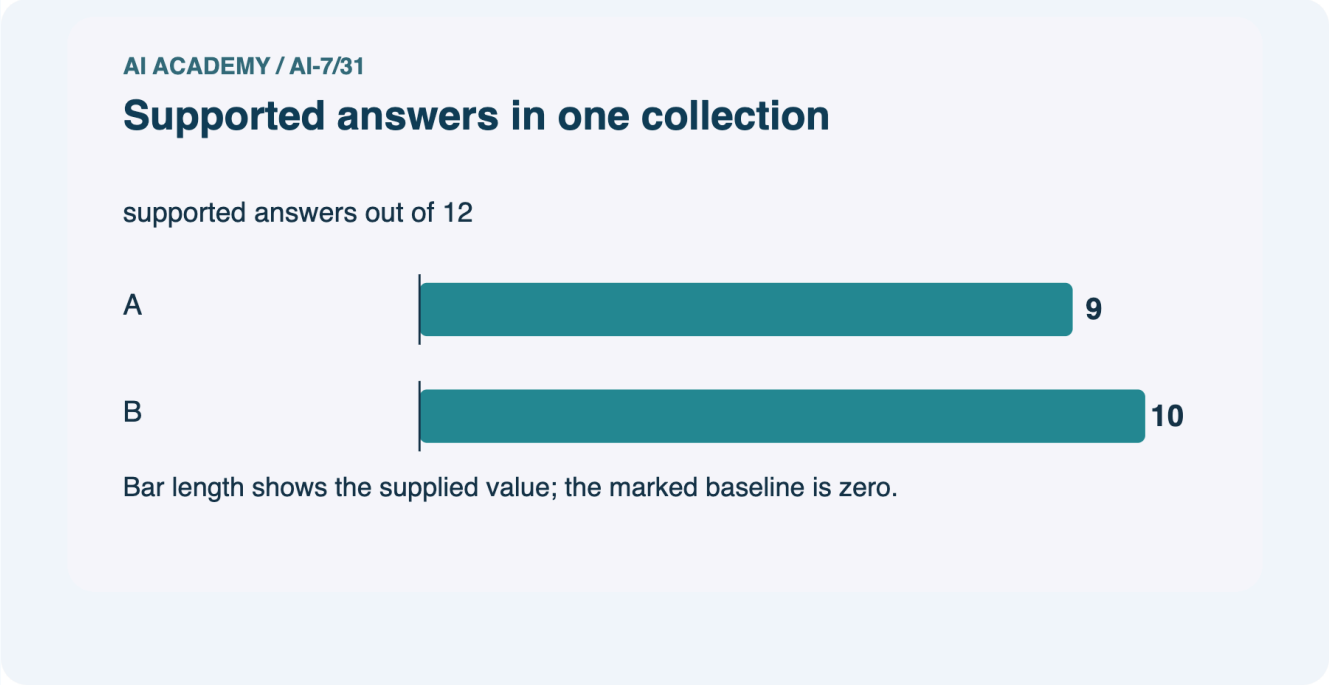
Follow the labelled picture with the matching explanation. Open the vector diagram

Report the **observation**: A is 75%; B is about 83.3%, a one-case improvement. → Link to the method: Name the collection, rubric and twelve-case denominator. → State limits: Do not infer that children learn more from this answer-support measure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Supported answers in one collection



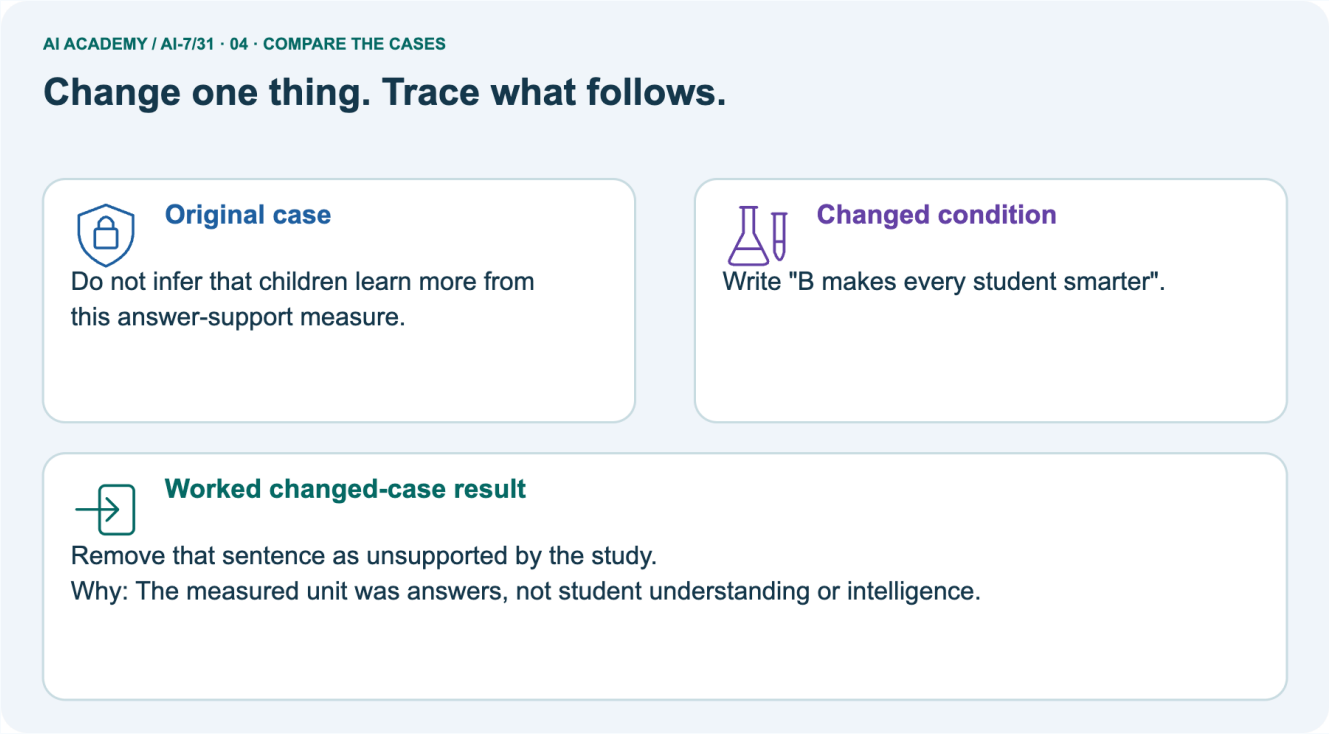
Follow the labelled picture with the matching explanation. Open the vector diagram

Supported answers in one collection. Units: supported answers out of 12. A: 9. B: 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

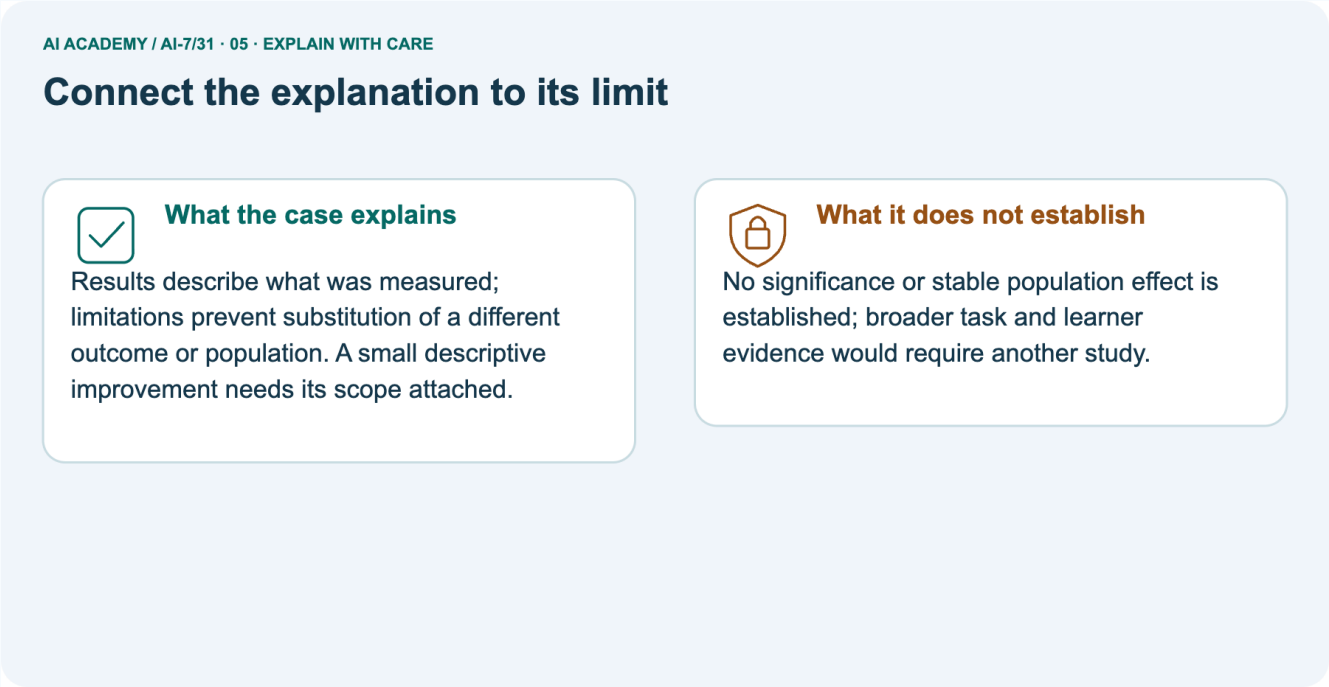
Original case: Do not infer that children learn more from this answer-support measure. Changed condition: Write "B makes every student smarter". Expected result: Remove that sentence as unsupported by the study.

Reason: The measured unit was answers, not student understanding or intelligence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

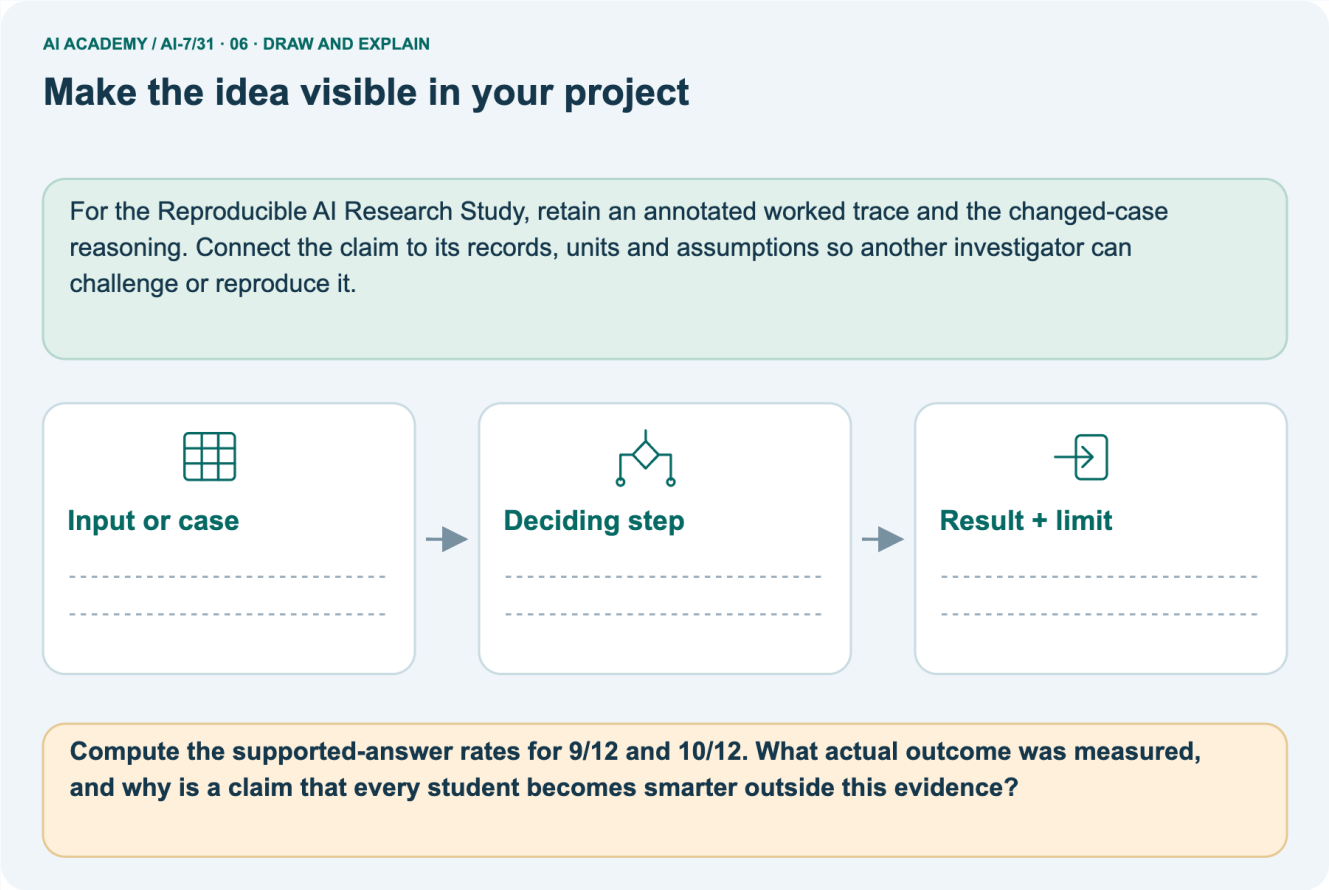
Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

Limit: No significance or stable population effect is established; broader task and learner **evidence** would require another study.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Writing results and limitations: the case

02. Follow the deciding steps

03. Supported answers in one collection

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.